

Engineering Graduation Project · MEF (GDDE) & ITC

# Cambodia Daily Consumer Price Index (CPI) Pipeline & Inflation Nowcasting

An automated data pipeline to scrape, clean, monitor, and nowcast daily inflation in Cambodia across all 12 UN COICOP divisions.

## PRESENTER

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## HOST INSTITUTION

**Ministry of Economy and Finance (MEF)**

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# What is the Consumer Price Index (CPI)?

## Measuring Cambodia's Cost of Living

- **What is CPI?** The primary economic indicator measuring changes in the cost of living and purchasing power over time.
- **The 12 Consumption Categories:** Tracks the full United Nations COICOP basket: Food & Non-Alcoholic Beverages, Housing & Utilities, Health, Transport, Communication, etc.
- **Measuring Inflation:** When the total cost of this basket rises, it reflects **inflation**—directly impacting everyday household expenses.

## Why High-Frequency CPI Matters

- **Ministry of Economy and Finance (MEF):** Essential for calibrating minimum wages, government budgets, and targeted social safety nets for vulnerable families.
- **National Bank of Cambodia (NBC):** Used to monitor macroeconomic stability, manage currency liquidity, and protect the value of the Khmer Riel.
- **Timely Decision Making:** Policymakers need high-frequency data to detect price shocks early rather than waiting months for retrospective reports.

# Problem Statement: The Official NIS Publication Delay

The Problem

### ⚠️ 30 to 60 Days Publication Lag

- **Manual In-Person Surveys:** The official CPI from the National Institute of Statistics (NIS) relies on manual survey enumerators visiting physical markets once a month.
- **Severe Reporting Lag:** Collecting, compiling, and auditing manual records takes **30 to 60 days** after month-end before official figures are released.
- **Retrospective Blind Spots:** By the time the official NIS report is published, ground economic conditions and price shocks have already played out.

### ⚠️ Cambodia's Vulnerability & Dual Currency

- **Extreme Expenditure Concentration:** Over **74% of the national basket** is concentrated in volatile Food (44.8%) and Transport/Fuel (12.2%), which change daily.
- **Dual-Currency Reality (USD & KHR):** Retail stores quote in both US Dollars and Khmer Riel; daily exchange rate movements alter real shelf prices instantly.
- **Delayed Policy Assistance:** Without real-time tracking, government relief interventions arrive too late to protect vulnerable populations.

# Project Objectives: What We Built

## 1. Build a Data Infrastructure

Deploy an enterprise PostgreSQL 16 Medallion data warehouse (Bronze, Silver, Gold schemas) to securely ingest, clean, and store high-frequency economic price data.

## 2. Automate Scraping with Airflow

Orchestrate 20+ automated web scrapers using Apache Airflow 2.9.3 to harvest >31,000 price quotes every morning at 02:00 ICT across all 12 COICOP divisions.

## 3. Monitor Scraped Data & Quality

Establish real-time observability dashboards in Metabase to continuously monitor scraper yields, store availability, unit normalization, and extreme price shifts (>20%).

## 4. Apply Machine Learning to Nowcast Inflation

Deploy ML models (momentum tracking & expanding-window regression) to nowcast current-month inflation in real time, eliminating the 30-day official NIS reporting lag.

# Methodology: Simple End-to-End Pipeline

Architecture

1. SCRAPE

 **Web Ingestion**

20 automated crawlers harvest daily prices from AEON, DeliShop, Chip Mong, Lucky, Community Pharmacy, redBus, BookMeBus, MOC Fuel, and official MEF FX rates.

2. BRONZE

 **Raw Landing**

Stores immutable atomic listings in PostgreSQL (`bronze.raw_prices`) with execution timestamps, barcodes, and raw JSONB payloads for audits.

3. SILVER

 **Clean & Classify**

Cleans text, converts USD to KHR, computes normalized unit prices (per kg/L to catch shrinkflation), matches items using vectors, and sorts products into 12 COICOP divisions.

4. GOLD

 **Indices & Marts**

Calculates daily Jevons geometric price indices, aggregates with official NIS expenditure weights into Headline & Core CPI, and feeds the daily ML nowcaster.

# Current Work: Pipeline Status & ML Inflation Nowcasting

Progress & ML

## 1. Automated Medallion Pipeline is 100% Operational

Apache Airflow triggers daily at 02:00 ICT, collecting >31,000 quotes across 20 sources and updating Bronze, Silver, and Gold schemas.

✓ 100% Live & Scheduled

## 2. Operational Health & Data Quality Monitoring

Metabase dashboards monitor scraper success rates (99.1%), daily price shifts, store volumes, and MEF FX freshness in real time.

✓ 100% Deployed

## 3. Machine Learning Inflation Nowcasting Engine ([ml/nowcaster.py](#))

Combines Month-to-Date observed daily facts with 7-day leading momentum in Food & Transport and festival shock adjustments to project month-end CPI with dynamic 95% confidence bands.

⚡ Active Model Evaluation

## 4. Out-of-Sample NIS Benchmarking & Thesis Defense Preparation

Evaluating nowcasting accuracy against historical official NIS benchmark releases; preparing final presentation for defense on September 30, 2026.

📝 Final Review